Team 25: NutriCart Hub

Design Prototype documentation: Design Process Overview + Ethical Considerations

Team member	Role	Student number
Aman Kushwaha	Project Management Lead	48781392
James Olbes	Design & Prototyping Lead	47449631
Anujitha Fernando	Technical & Dev Lead	48646848
Kenan Chu	Documentation & Theory Lead	46413596

GitHub repository: https://github.com/Anujithaaa/DECO3500-Team25-NutriCart
WIKI: https://github.com/Anujithaaa/DECO3500-Team25-NutriCart/wiki/

Where to find everything in the repo

What	Where
README (start here: run steps, quick links, repo map)	github.com/Anujithaaa/DECO3500-Team25-NutriCart
Prototype code (Home Hub + Cart Unit, Pico LED)	github.com/Anujithaaa/DECO3500-Team25-NutriCart/tree/main/prototype/app-v2
WIKI: Design Process Overview, Ethics, User Research, Evaluation Plan, Stand-ups	github.com/Anujithaaa/DECO3500-Team25-NutriCart/wiki/
Research records (de-identified interviews, shadowing)	github.com/Anujithaaa/DECO3500-Team25-NutriCart/tree/main/documents/research
Evaluation: G-01, G-02, tradeshow feedback, iteration evaluation	github.com/Anujithaaa/DECO3500-Team25-NutriCart/tree/main/documents/evaluation
Design artefacts (concepts, storyboards, mock-ups)	github.com/Anujithaaa/DECO3500-Team25-NutriCart/tree/main/documents/design
Poster, pitch script, promo video, try-it card, this PDF	github.com/Anujithaaa/DECO3500-Team25-NutriCart/tree/main/documents/promo
Prototype photos (Hub, Cart Unit, tradeshow)	github.com/Anujithaaa/DECO3500-Team25-NutriCart/tree/main/documents/prototype%20photos
Task board (GitHub Projects, all 30 issues)	github.com/users/MaxCoder07/projects/1
Team charter and planning	github.com/Anujithaaa/DECO3500-Team25-NutriCart/tree/main/documents/team

Pages 2 onward are PDF exports of the WIKI pages *Design Process Overview* (~1,000 words excluding references) and *Ethical Considerations*.

Design Process Overview

Poster: PDF · **Promotional material:** pitch script · **Prototype:** runs on a laptop and phones, no login; run steps for app-v2 and v0.3 are in the README. · **Team 25:** Aman Kushwaha, James Olbes, Anujitha Fernando, Kenan Chu

Summary

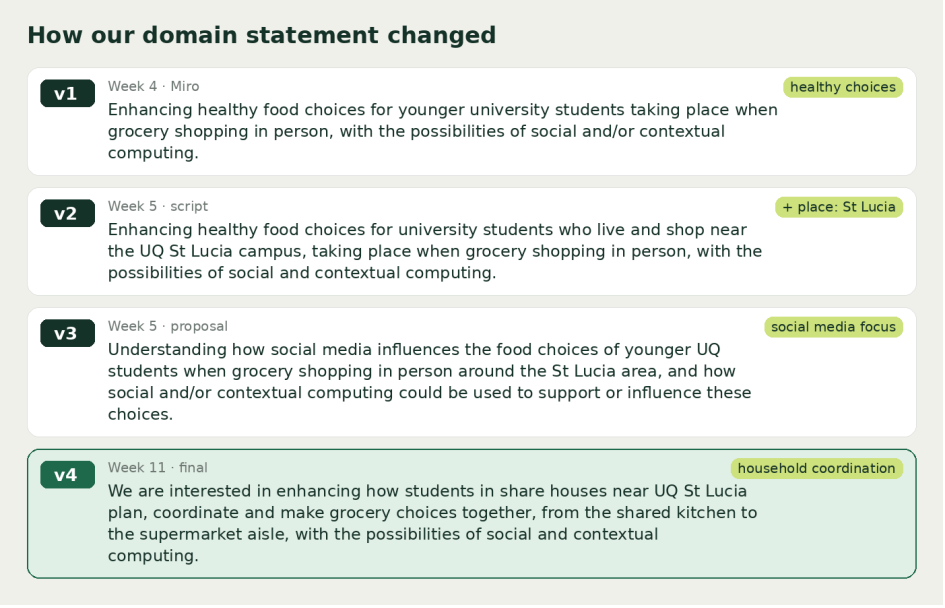
We investigated how students in share houses near UQ St Lucia plan and coordinate groceries. Price decides most purchases, and “healthy” means something different to everyone. Most shopped alone, yet shopping overlapped through shared staples, fridge space and lists sent to whoever was going. The useful social influence isn’t a feed of friends’ meals but household coordination: who needs what, who’s going, what’s in the fridge, and who pays.

NutriCart Hub is our response. Housemates plan together at the kitchen Home Dock, and whoever shops takes the Cart Unit with the household's list, requests and budget. The social computing sits in six moments where several housemates take part (G1–G6).

Process

1. Understanding the domain (Weeks 3–6)

Our first domain statement was about "enhancing healthy food choices". Round 1 interviews showed **price first** ("99 out of 100 times") and social influence through **trust and timing**: a housemate who finds the cheap milk. Food-choice tools are mostly individual (Marshall et al., 2022), and social media inspiration rarely carries into the store (Wang et al., 2024). Proposal feedback asked: whose definition of healthy? Why an app?



Four versions of our domain statement, from "healthy choices" to household coordination.

2. An app as a probe, not a product (Weeks 6–7)

v1, a social shopping app with friend recommendations, group goals and recipes, was a **design probe**. People wanted **prices**, not friends' recipes, and one read a healthy-dinner goal as "essentially telling me that I'm fat".

3. Reframing and divergence (Weeks 7–8)

Synthesis reframed the problem: *support real grocery decisions without adding effort, judgement or another app.* Non-app concepts (a budget buzzer, a "grocery baton") combined into **v2**: a cardboard Home Dock and detachable Cart Unit, tested with Wizard of Oz. Testing kept the **shared list, running budget and neutral compare**, and told us to keep alerts rare ("noise, not help") and cut data entry. Our four requirements: inform without judging, price first, no extra effort, and support coordination the household already does.



Storyboard: with the Hub, the household's needs travel with the shopper. More concepts in documents/design.

4. "What's the point of the social part?" (Week 9 Stand-up)

Staff said the social aspect was weak, so we redesigned around **six group moments** on the time–space matrix:

	Same time	**Different time**
Same place	G1 plan together at the Dock · G5 Split Shop in store	G2 leave a request · G3 hand over the baton · G6 return and settle
Different place	G4 one quick question home	–

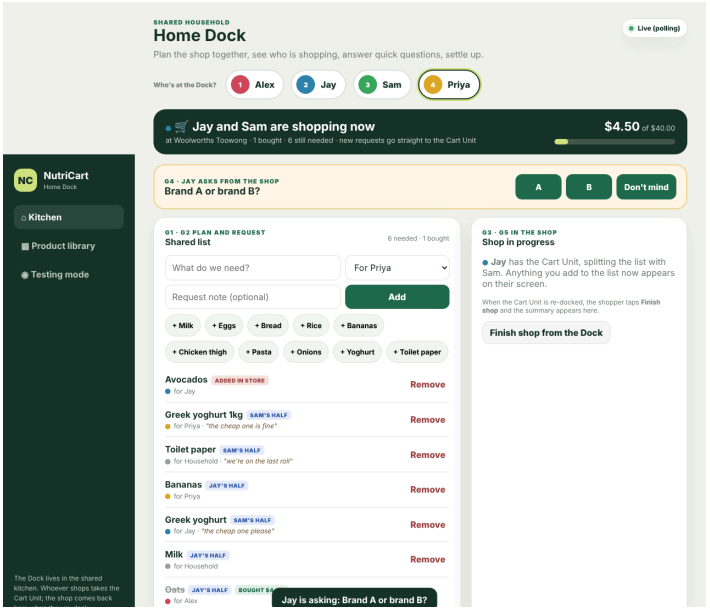
The Hub now **facilitates** coordination households already do (Olsson et al., 2020), uses visible traces rather than notifications (Schmidt, 2002) and gives every housemate something back (Grudin, 1994).

The Designing for Location Dependence framework (Design for Location, n.d.) showed that decisions start at home, so NutriCart follows a home → store → home trajectory: the Dock belongs to the kitchen; the Cart Unit shows only what matters in store.

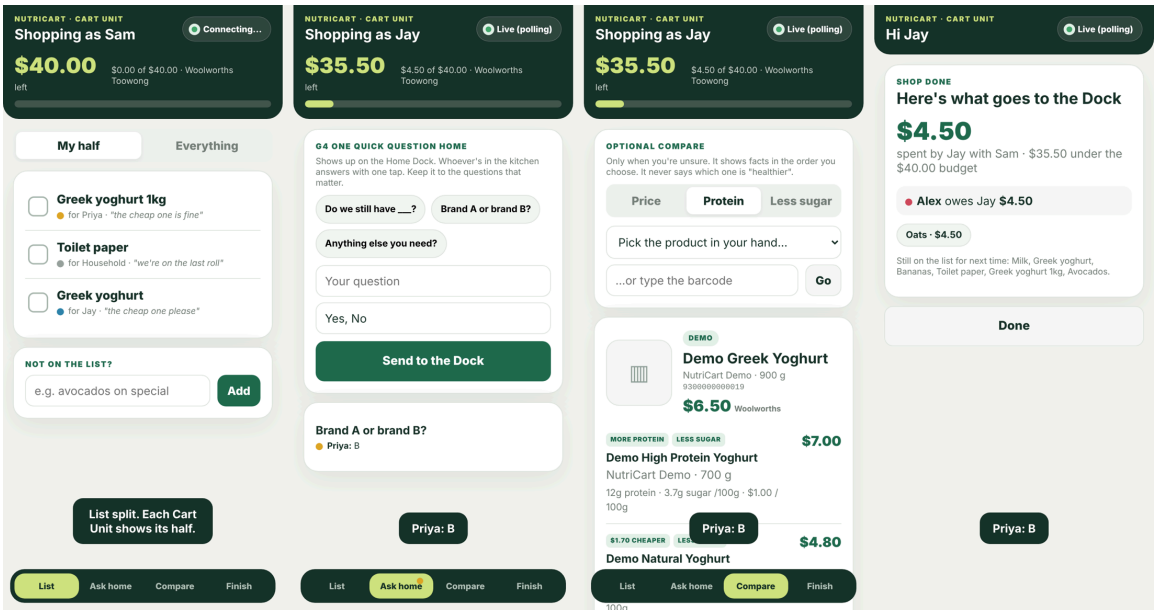
5. v3: multi-user and live, going physical (Weeks 10–11)

v3 is an **integration prototype** (Houde & Hill, 1997), real where the social experience depends on it. James built a live-sync backend and an Expo Cart Unit with **barcode scanning**; v0.3 added requests,

"Jay is shopping now", Ask home, two-phone Split Shop and a cost split. Prices are entered by hand, and compare orders products by **the shopper's** priority, never "healthier".



v3 Home Dock during a live shop.

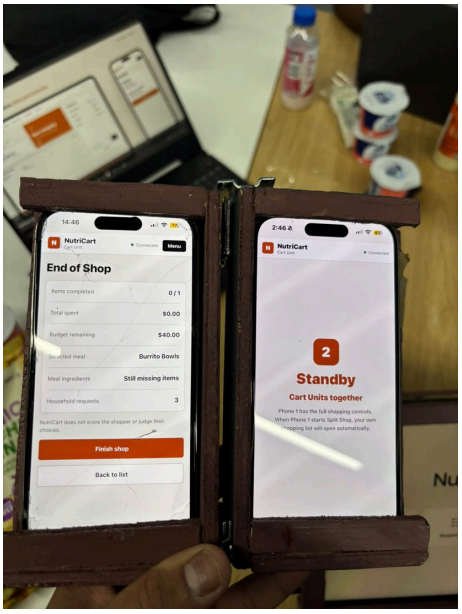


v3 Cart Units: Split Shop, Ask home, compare and the return summary.

James built physical housings; Kenan's **app-v2** redesigned the Hub and Cart Unit, the version we tested and showed at the tradeshow.



Home Dock housing with two Cart Unit mounts.



Split Shop on two Cart Units.

6. Evaluating with groups, in context (Week 12)

Two group sessions with four students who live together, one in their own kitchen (Evaluation Plan). All four valued the budget, Compare or deals most; all struggled with first use (F2). The sharpest finding: **the group chat is our real competitor (F3)**. G-01 called requests "another group chat" and felt the design was “promoting the opposite of being social”, while G-02 valued seeing what was already at home. The social value sat in the live, in-store moment, so we will lead with same-time coordination, work alongside the chat, simplify first use, and swap the LEDs for a chime.

7. Tradeshow (Week 12)



Visitors at our stall (faces blurred).

Visitors used the Dock and two Cart Units in pairs, and five teams left feedback on Miro (summary). They confirmed the two-place design ("it splits the experience across two distinct physical environments", Team 18). Three problems repeated:

- **First use** is hard: three teams asked for a guide (F2).
- **Unlabelled numbers** in the meal editor, deals and scanning not fully working.
- **The group chat**: "instead of replacing group chats, integrate with it" (Team 18; F3).

Next: a first-use walkthrough, clear labels, working compare, sharing to the chat, and a private cost split (Ethics E1). James's iteration evaluation sets out the next test.

Limitations

Samples were small and mostly people we could reach, and we consulted no domain expert (e.g. student-housing staff); studio critique stood in. Final testing (two short group sessions and tradeshow peers) shows first use, not how a household behaves over weeks (Grudin, 1994). v2 was Wizard of Oz, and v3 supports one household per server.

Who did what

Member	Main contributions
Aman Kushwaha (Project Management Lead)	Plan, charter, task board, stand-ups; synthesis; de-identification; evaluation plan; pitch and tradeshow pack
James Olbes (Design & Prototyping Lead)	Shadowing; physical prototypes (v2, v3 housings); backend and Expo Cart Unit; iteration evaluation
Anujitha Fernando (Technical & Dev Lead)	GitHub repo set-up; integration and multi-device testing of v3; poster layout and printing
Kenan Chu (Documentation & Theory Lead)	Physical prototypes; interview synthesis; personas and scenarios; research archive and README; app-v2 redesign and frontend; WIKI and final PDF

We shared 23 research and testing sessions; the evidence register shows who ran each.

Use of AI: every member used Claude and ChatGPT; see the README.

References

- Design for Location. (n.d.). *Designing for location dependence: A framework*. https://designforlocation.org/framework.html
- Grudin, J. (1994). Groupware and social dynamics: Eight challenges for developers. *Communications of the ACM, 37*(1), 92–105.
- Houde, S., & Hill, C. (1997). What do prototypes prototype? In M. Helander, T. Landauer & P. Prabhu (Eds.), *Handbook of Human-Computer Interaction* (2nd ed., pp. 367–381). Elsevier.
- Marshall, J., Jimenez-Pazmino, P., Metoyer, R., & Chawla, N. V. (2022). A survey on healthy food decision influences through technological innovations. *ACM Transactions on Computing for Healthcare, 3*(2), Article 25. https://doi.org/10.1145/3494580
- Olsson, T., Jarusriboonchai, P., Woźniak, P., Paasovaara, S., Väänänen, K., & Lucero, A. (2020). Technologies for enhancing collocated social interaction: Review of design solutions and approaches. *Computer Supported Cooperative Work, 29*, 29–83.
- Schmidt, K. (2002). The problem with "awareness". *Computer Supported Cooperative Work, 11*(3–4), 285–298.
- Wang, C.-H. A., Sher, S. T.-H., & Chung, C.-F. (2024). From viral content to real-life cuisine and beyond: Examining teenagers' interactions with TikTok food videos and the influence on their food practices. *Proceedings of the ACM on Human-Computer Interaction, 8*(CSCW2), Article 389. https://doi.org/10.1145/3686928

Ethical Considerations

We assessed NutriCart Hub with the **Ethical Disclaimer** from the Ethics for Designers toolkit (Gispen, 2017), because it asks designers to state who is affected, what could go wrong and what they take responsibility for, which suits a tool that sits inside a shared home. The disclaimer asks a team to describe the situation, list everyone affected, state its intentions, imagine how the design could cause unethical outcomes for each stakeholder, and decide openly which of those outcomes the team takes responsibility for. Following the course's emphasis on ethical design thinking throughout a project (Loke & Matthews, 2020), this is a **living page**: we revisit it at each milestone (section 6).

1. The current situation

Students in share houses near UQ St Lucia coordinate groceries informally: group chats, remembered requests, a list on someone's phone. Our research found price dominates their choices, "healthy" means different things to different people, and most people shop alone even when they live with others. Coordination problems are real (doubles, forgotten requests, awkward cost sharing), but so is sensitivity around money, food and bodies.

2. Who has a stake

Direct	Indirect
The housemate who shops (uses the Cart Unit)	Partners, guests and visitors who see the Dock in the kitchen
Housemates at home (use the Dock)	Supermarket staff and other shoppers (in-store use and testing)
A housemate on a tighter budget	Retailers whose prices appear in the product library
A housemate who doesn't share food or doesn't want to take part	Open Food Facts (receives barcode lookups)
Housemates with dietary, cultural, religious or allergy needs	Expo (the Cart Unit runs in the Expo Go app)
Someone with a difficult relationship with food or their body	Research participants and the team

3. Our intentions

NutriCart Hub is for **student share houses**. It is used **at the kitchen bench and in the supermarket aisle**. Its purpose is to **make the coordination housemates already do easier and fairer**: who needs what, who is shopping, what's already at home, and how to split the cost. It is not meant to change what anyone eats or to rate their choices.

The risks below are written for the full design (v0.3 in the repo). app-v2, the version we tested in Week 12, has a subset of its features: it has no cost split, so E1 and E5 apply when the split returns.

4. Imagining how it could go wrong

#	Stakeholder	How the design could harm them	What we changed or decided	Status
E1	Housemate on a tighter budget	The **cost split** and running total make income differences visible and turn shared shopping into a tally of debts. One participant said shared goals could be "stressful for one person because if he's getting less pay."	Shared staples are split evenly by default; personal requests are charged to whoever asked. Prices are optional, and unpriced items are left "to settle by hand". The summary can be dismissed without settling. *Next:* soften "owes" wording; let a household choose "don't split". **Tradeshow:** a peer worried that showing the budget and who owes what "might be a bit embarrassing for people who are tighter on money, and guests could see it too" (Team 18); another warned that seeing who spends what "could cause arguments". *Next:* a private settle-up shown only to the people involved, and hide totals when the Dock is idle.	Partly addressed; concern confirmed by peers
E2	Someone with a difficult relationship with food or their body	Group health goals and comparison can feel like judgement. A participant reacted to a dinner goal with: "It's essentially telling me that I'm fat... if anything, it could cause body dysmorphia." (Others found goals motivating, so reactions were mixed.)	v3 has **no group health goals, leaderboards, calories or health scores**. Compare is optional, shows facts only, and is ordered by the shopper's chosen priority. It never says "healthier".	Addressed by design
E3	Every housemate	**Awareness becomes monitoring** (Schmidt). Showing who added what, who is shopping and what they bought could feel like surveillance inside the home.	The Dock shows that someone is shopping and the store they chose. There is no live location tracking. Aliases are fine. **Honest limit:** past shops, their cost splits and barcode scans (with the shopper's alias) stay on the server until the next reset. *Next:* auto-delete past shops and scans after settling. **Tradeshow:** peers named spending and dietary preferences as what housemates may not want shared, and asked for control over what is shown; one team noted we already show "shopping activity without tracking users' locations" (Team 20). *Next:* per-item "just for me" privacy.	Partly addressed
E4	The housemate who ends up maintaining the Dock	**Unequal work**: one person does the adding and organising while everyone benefits (Grudin's work vs benefit disparity), and this invisible labour often falls on the same person.	Quick-add chips, a meal → "Add what's missing" button, requests that take one tap. Testing confirmed the upkeep feels "super tedious" (G01-P2); the finished shop now updates what's at home.	Confirmed in testing; partly addressed

#	Stakeholder	How the design could harm them	What we changed or decided	Status
E5	Housemates who don't share food, or don't want to take part	The design's **script** (Akrich) assumes a cooperative household; non-participants may feel pressured to join or be left out. **In v3, items marked "for the household" are split across every token, so a housemate who never uses NutriCart still appears as owing a share.**	Everything works for one person, and personal requests are only charged to whoever asked. *Next:* let a housemate opt out of shared costs, and remove unused tokens.	Partly addressed
E6	Housemates with dietary, cultural, religious or allergy needs	A product library and meal board built by our team reflect **our** food habits.	Products are editable in the Product library; compare shows the shopper's priority, not ours. Meals are currently fixed demo meals. *Next:* add and edit meals. *Next:* allergy and dietary notes per housemate visible on requests.	Open
E7	Housemates with low vision or colour-vision differences	Colour-only tokens exclude some users.	Tokens carry a **number and a name** as well as a colour; large touch targets. *Next:* contrast check.	Partly addressed
E8	Retailers, the household	**Commercial capture**: a compare screen is an obvious place to sell rankings to brands. Scraping prices could breach retailer terms.	Ordering is only by the shopper's chosen priority; no sponsored ordering, ever. Prices are entered by hand from observed shelf prices; **nothing is scraped**.	Decided
E9	Everyone using it	**Data**: shopping lists reveal diet, budget and routines. Barcode lookups go to Open Food Facts. We decided not to host the demo, so the database stays on our own laptop.	No accounts, no analytics, aliases only. The server stores list items, past shops with their cost splits, questions and barcode scans until the next reset. The demo database is reset between sessions. Database files are git-ignored.	Addressed for the prototype
E10	Other shoppers, store staff, research participants	**Research harms**: shadowing felt intrusive; filming in store captures bystanders.	Moved from shadowing to lower-burden sessions; consent script; manager permission in store, or a simulated shop; no faces or bystanders; recordings and transcripts kept off public GitHub, deleted at semester end.	Addressed
E11	The shopper	**Dependence**: if the server or connection fails mid-shop, the list disappears.	Automatic fallback from live connection to polling; hotspot setup. **Tradeshow:** a peer described poor reception in the supermarket leading to an item bought twice and a wrong budget (Team 16). *Next:* keep a cached copy on the Cart Unit and show when it is out of date.	Partly addressed

5. What we take responsibility for, and what we don't

We take responsibility for: E1–E11 as far as the design can reach. These are consequences of choices we made (what's shown, to whom, how costs are split, what's stored, how we ran research). Winner (1980) reminds us that a detail like "split shared items evenly" quietly settles a social question for the

household. So the split rule is visible and explained, not hidden, and making it optional is our next step (E1).

Outside our control, but named: existing conflict between housemates (the Dock can surface it but not resolve it); retailer pricing accuracy; a household choosing to use the cost split to pressure someone. We can't prevent these, so we don't design features that amplify them, such as rankings, reminders to pay, or debt history.

6. Revision log

Date	Milestone	What changed on this page
Wk 5	Proposal Q&A and peer comments	Q&A raised: what is "healthy" for everyone; dietary, religious and cultural needs; eating disorders. Peer comments raised privacy and user control.
Wk 7–8	Prototype interviews (v1, v2)	Evidence for E1 (pay) and E2 (body image) → group goals made optional; later left out of v3
Wk 8	Shadowing attempt	E10: method changed
13 Oct	v3 build	E3, E8, E9, E11 added; first full disclaimer
14 Oct	Research archive	Transcripts removed from the repo and its history; all sessions re-published as de-identified records; raw files kept privately on OneDrive (E10). Lesson: de-identify before anything leaves the private folder.
19 Oct	Evaluation round 1 (G-01, G-02)	**E4 confirmed:** keeping Already at home updated "seems super tedious" (G01-P2), so the finished shop updates it automatically (v0.3) and one-line quick-add is next. **E5 strengthened:** G-01 would rather use the group chat, so NutriCart must work alongside the chat instead of pulling the whole house onto it. **E1 and E3 not yet tested** (app-v2 has no cost split; the privacy question wasn't recorded), so we ask tradeshow visitors.
20–22 Oct	Tradeshow	Peer feedback from five teams (summary). **E1 confirmed** (visible budgets and debts could embarrass a housemate on less money, or be seen by guests). **E3 refined** (spending and dietary preferences are the sensitive data; location is not shown). **E5 echoed** (a housemate who won't use the Hub; one person always paying). **E11 confirmed** (poor reception in store). Next steps added to each row.

References

- Akrich, M. (1992). The de-scription of technical objects. In W. E. Bijker & J. Law (Eds.), *Shaping technology/building society* (pp. 205–224). MIT Press.
- Gispen, J. (2017). *Ethics for Designers* toolkit: Ethical Disclaimer. ethicsfordesigners.com
- Grudin, J. (1994). Groupware and social dynamics: Eight challenges for developers. *Communications of the ACM, 37*(1), 92–105.
- Loke, L., & Matthews, B. (2020). Scaffolding of interaction design education towards ethical design thinking. In G. Melles (Ed.), *Design thinking in higher education* (pp. 165–181). Springer. https://doi.org/10.1007/978-981-15-5780-4_8
- Schmidt, K. (2002). The problem with "awareness". *Computer Supported Cooperative Work, 11*(3–4), 285–298.
- Winner, L. (1980). Do artifacts have politics? *Daedalus, 109*(1), 121–136.